

GENERAL SPECIFICATION G7.2

Mechanical Ventilation Systems

CONTENTS

G7.2 Mechanical Ventilation Systems1

 G7.2.1 General.....1

 G7.2.2 Reference Standards.....1

 G7.2.3 Submissions by the Contractor2

 G7.2.4 Ventilation Fans3

 G7.2.5 Ductwork.....8

G7.2 MECHANICAL VENTILATION SYSTEMS

G7.2.1 General

- (a) The Contractor shall be responsible for detailed design of the ventilation systems for all plant areas as indicated on the Drawings and as specified.
- (b) The values of resistance to air flow of items of equipment, ductwork and all other items of the system shall be determined by the Contractor. Any details shown on the Drawings and included in the Specification are notional and shall be verified by the Contractor.
- (c) Locations of fans, ductwork and other items shall follow the general layout indicated on the Drawings prepared by the S.O. as far as possible, but the Contractor shall prepare detailed layout drawings to suit the ventilation requirements of the particular area and the layout of the Works.
- (d) The Contractor shall submit to the S.O. for approval, calculations and drawings for the proposed systems. The drawings shall detail all fans, flanges, methods of construction, locations of splitters, dampers, louvres, bends, air filters, supports, brackets, access panels and test holes. All air inlets shall incorporate filters, materials and protection systems consistent with the requirements of the Specification.
- (e) The Contractor shall submit samples of ductwork, fittings, materials and finish for the S.O.'s approval.
- (f) Manufacture shall not commence until the details of the proposed ventilation systems have been approved by the S.O..

G7.2.2 Reference Standards

- (a) Unless otherwise specified, ductwork and fittings shall comply with the relevant Reference Standards listed below:

Standard	Subject
SS 553	Code of practice for Mechanical Ventilation and Air Conditioning in Buildings
Fire Code	Code of Practice for Fire Precautions in Buildings 2018
BS 4994	Specification for the design and construction of vessels and tanks in reinforced plastics.
	Code of Practice on Environmental Health
SS333	Specification for Fire Dampers
BS 3757	Specification for rigid PVC sheet.
BS 5315	Specification for hose clamps.
BS 476 Part 7	Method of classification of the surface spread of flame of products.

Standard	Subject
SAI-SAA AS 2634	Chemical Plant Equipment Made from Glass-Fibre Reinforced Plastic (GRP) Based on Thermosetting Resins.
BS 4675 Part 1.	Mechanical Vibration in Rotating Machinery
BS 476 part 8	Fire Test on Building Materials and Structures
BS 476 Part 5	Method of test for ignitability
BS EN ISO 5135	Methods for testing and rating air terminal devices for air distribution systems

- (i) American National Bureau of Standards: Voluntary Product Standard PS-15-69 for construction of fibreglass ducts.
- (ii) Sheet Metal and Air-conditioning Contractors National Association (SMACNA): HVAC Air duct leakage manual.
 - SMACNA : Thermoplastic Duct Construction Manual
 - SMACNA : Manual of Fibre Glass Duct Construction Standards
 - SMACNA : Manual of Duct and Sheet Metal Construction for Ventilating and Air Conditioning Systems
- (iii) American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE): Hand book

G7.2.3 Submissions by the Contractor

Contractor shall submit the following:

- (a) Drawings
 - (i) Fully dimensioned shop drawings;
 - (ii) Standard Detail drawings of standard items including details of material to be used full dimensions etc; and
 - (iii) Detail drawings of special items including details of the materials to be used, full dimensions etc.
- (b) Data
 - (i) Calculations of air flow, pressure drop, structural strength, acoustic levels for the components and system offered;
 - (ii) Catalogues and data sheets, including temperature/pressure ratings for ducts, fittings and joints;
 - (iii) Method Statements for Installation;
 - (iv) Methods of support; and
 - (v) Methods of control of line and level of ductwork during installation.

- (c) Method and programme of ductwork and fittings testing and cleaning: The Contractor shall submit for the approval of the S.O., details of his proposed methods and programme for ductwork testing and cleaning and for adjustment of airflow by air flow measurement and setting of dampers.
- (d) Structural support of ductwork: The Contractor shall submit for the approval of the S.O. full details and structural calculations, for the proposed supports for ductwork above ground level to the levels and routes proposed. These shall include, where required wind loadings, foundation details and loadings, structural metalwork details and loadings, and shall include all details of codes and standards referred to.
- (e) Test Certificates.
- (f) Manufacturer's quality assurance/quality assurance program.

G7.2.4 Ventilation Fans

- (a) General
 - (i) The Contractor shall verify values of the resistance to air flow of items of equipment, ductwork and/or the total distribution system based on the equipment offered and to provide fans capable of delivering the required air volume when operating against the actual total system resistance. All fans with nominal rating of above 7.5 kW shall have a minimum fan total efficiency of 75%.
 - (ii) All fans shall be so constructed to withstand continuous operation with minimum vibration and noise. The fans shall be designed to deliver the specified air volume at the prevailing site conditions.
 - (iii) All fans shall be type' tested in accordance with the requirements of BS 848 and shall be selected to give the air volume flow rates and sound power levels specified.
 - (iv) All fans shall be constructed to a fully developed design and shall be capable of withstanding the pressures and stresses developed during continuous operation at the selected duty. Additionally, all belt-driven fans shall be capable of running continuously at ten per cent more than the maximum selected duty speed.
- (b) All fans shall be fitted Anti-vibration mountings:
 - (i) Fans which serve as part of an engineered smoke control system, shall be suitable for operation at an ambient temperature of 250°C for a period of 2 hours. The motors for such fans shall fully meet the requirements of the Singapore Fire Code.
 - (ii) The speed of all ventilation fans shall not exceed 1500 revs/min except where specifically identified and approved by the S.O.. The noise level generated by the ventilation system shall not exceed the specified level in any case. If additional noise abatement measures are required, the same shall be included in the design and installation at the contractor's own costs.

- (c) Manufacturer
 - (i) All fans and their respective drives shall be the product range of single manufacturer experienced in the design and construction of similar equipment and who has been manufacturing fans of similar duties for the last ten years. In addition, all guarantees, test certificates etc. shall be deemed to apply to the entire assembly.
 - (ii) The Contractor shall submit details of the make and type of fan with indication of sound power level and fan total efficiency at the selected duty for the approval of the S.O.. Where fans are supplied with noise attenuators, full details of the attenuators shall be provided.
 - (iii) Fan drives shall be designed to deliver the specified air volumes under the local site ambient temperature and relative humidity with sufficient allowance for normal fluctuation so that no overload occurs.
 - (iv) Fans shall be installed using bolts, nuts and washers with all 'as cast' bearing surfaces for bolt heads and washers counter faced. Holding-down bolts for fans and motors shall be provided with means to prevent the bolts turning when the nuts are tightened. Fans heavier than 20 kg shall be provided with eyebolts or other purpose made lifting facilities.
 - (v) Methods of support must be submitted.
 - (vi) Method statements for installation and maintenance must be submitted.
- (d) Impeller and shaft: The shaft and impeller assemblies of centrifugal, axial flow and mixed flow fans shall be statically and dynamically balanced. All propeller fans shall be statically and dynamically balanced where the impeller diameter is 250 mm or greater. Where indicated, limits of vibration severity shall be in accordance with BS 4675 Part 1.
- (e) Bearings
 - (i) Fan bearings shall be of a type suitable for the installed altitude of the fan. They shall be grease/oil ball and/or roller type or alternatively oil-lubricated sleeve type. All bearing housings shall be precision located in position and arranged so that bearings may be replaced without the need for realignment. Bearing housings shall be protected against the ingress of dust and, where fitted with greasing points, they shall be designed to prevent damage from over-greasing.
 - (ii) For grease lubricated systems the bearings shall be provided with grease in amount and quality recommended by the bearing manufacturer. For oil lubricated systems the housings shall provide an adequate reservoir of oil and shall include a filling plug and be oil tight and dust proof. Systems other than total loss types shall include an accessible drain plug. All bearing lubricators shall be located to facilitate easy maintenance. Greasing points shall be easily accessible.
- (f) Vibration isolation: All dynamic machinery shall be isolated from the building structure by vibration isolators and/or vibration isolation materials which shall be purpose designed and selected to suit the machinery.

- (g) Sound control: The Contractor shall make every effort to contain noise at its source. Packaged sound absorbers such as attenuators, silencers, mufflers, etc. shall be fitted immediately adjacent to the fan so that the sound power is reduced at source. The fan sound pressure level, measured under installed conditions at a distance of three fan diameters from the fan centre, shall not exceed 85 dB(A) at 1m. If necessary a silencer may be provided and the cost shall be deemed to be included in the tender sum.
- (h) Flexible connections: All fans to be connected to a ductwork system shall be connected by means of flexible connections at the suction and discharge ends. The flexible connections shall be lead impregnated vinyl and shall be of approved materials and installed to allow replacement without the need to dismantle the ductwork. The flexible material shall be fixed onto the fan inlet and outlet such that no air leakage occurs.
- (i) Mounting and Finishes
 - (i) Fans shall be located generally as indicated on the Drawings. The fans shall be resiliently mounted on spring/neoprene vibration isolators. All support beams, supports legs, platforms, hangers and anchoring bolts required for the proposed installations as recommended by the fan manufacturer shall be provided by the Contractor.
 - (ii) Vee belts shall not be placed on drives, except for initial fitting, until the equipment is about to be commissioned. Belts which have taken a permanent set through remaining idle shall be replaced before hand over.
 - (iii) All fans shall be factory painted in accordance with the painting specifications specified elsewhere. Any damages to the finish that have occurred during transit, storage, installation or otherwise shall be made good as recommended by the manufacturer and to the satisfaction of the S.O.. Where fans cannot be touched up to the satisfaction of the S.O., they shall be re-painted over the entire external surface area.
- (j) Centrifugal Fans
 - (i) Centrifugal fans consuming more than 7.5 kW at the fan shaft shall be, unless otherwise stated, of the backward curved bladed type having a fan total efficiency of not less than 75%.
 - (ii) Casing shall be of the heavy galvanised sheet construction with adequate stiffening and bracing. It shall be free from vibration and drumming effect during normal operation. The fan casings shall be constructed to permit withdrawal of the fan impeller after fan installation. Fans shall be provided with flanged outlet connections and inlet connections. A valved and plugged drain point shall be fitted at the lowest point in fan casings. Permanent indication shall be provided to show the correct direction of rotation of the fan impeller. Fan casings shall be provided with removable access panels which shall incorporate purpose made air seals. The sizes of access panels shall be such as to facilitate cleaning and maintenance of the impeller.

- (iii) Impellers shall be of mild steel, aluminium alloy or plastics material, welded or other approved construction, with spiders or hubs of robust design. Impellers shall be rigidly fixed to the steel shafts that shall be adequately sized and proportioned to ensure that the maximum operating speed do not exceed 70% of the first critical speed.
 - (iv) The shaft shall be carried in ring lubricated self-aligning sleeve bearings for shafts of above 150 mm diameter. Each bearing shall be provided with a large oil capacity to ensure adequate and efficient lubrication. Shafts with diameter of 150 mm and below, grease lubricated self-aligning ball bearing resiliently mounted to reduce noise transmission shall be used. The bearings shall be equipped with grease escape valve to permit replacement of lubricant without dismantling the bearing housing.
- (k) Axial Flow Fans
 - (i) Axial flow fan casings shall be rigidly constructed of galvanised mild steel or aluminium alloy, stiffened and braced where necessary to obviate drumming and vibration. Mounting feet shall be provided where necessary for bolting to a base or to supports. Inlet and outlet ducts shall terminate in flanges to facilitate installation of flexible connections and easy removal. For in-duct mounting fans the length of the fan casing shall be greater than the combined length of the impeller(s) and motor(s) and electrical connections to the motors shall be through an external terminal box welded to the casing.
 - (ii) Impellers shall be of steel, aluminium or plastics and the blades shall be secured to the hub or the blades and the hub shall be formed in one piece. The hub shall be keyed to the shaft. Blades shall be aerofoil section or laminar and capable of pitch adjustment where indicated. For adjustable impeller blades, this adjustment shall be carried out without the removal of the wheel.
 - (iii) Where axial flow fans are driven by motors external to the casings of the fans the requirements for drives and guards as mentioned elsewhere in this Specification shall be met. Unless otherwise indicated, a guard is not required for any part of a drive which is inside the fan casing. An access panel with purpose made air seal shall be provided in the fan casing; the access panel shall be sized to facilitate maintenance.
 - (iv) Where axial flow fans of the bifurcated type are indicated, the motors shall be out of the airstream. Motors may be placed between the two halves of the casing in the external air or may be placed within the fan casing provided that effective ventilation is given to the motors.
- (l) Propeller Fans
 - (i) Propeller fans shall be ring mounted or diaphragm mounted as indicated. Impellers shall be of steel or aluminium and the blades shall be secured to the hub or the blades and hub shall be formed in one piece. The entire assembly shall be dynamically balanced.
 - (ii) Mounting plates and motor supports shall be adequately sized and constructed to provide a completely rigid support. Mounting plates shall be of hot dipped galvanised steel pre-formed with a bell mouth entry to maximise aerodynamic efficiency.

- (iii) Air inlet and outlet guard shall be provided to prevent against accidental contact with the propeller and to prevent the intrusion by animals, birds etc. The guards shall be manufactured of hot dipped galvanised steel rods and/or wire mesh.
- (iv) Where necessary or as specified, back draught dampers shall be provided to ensure only unidirectional airflow. The damper shall be opened/activated by the air pressure from the fan during fan operating and shall close under gravity when the fan stops. The damper shall be designed to ensure that damper rattling noise is non-present during or in between fan operation.
- (m) In-line Centrifugal and Mixed Flow Fans
 - (i) Mixed flow fan casings shall be rigidly constructed of mild steel, or aluminium alloy stiffened and braced where necessary to obviate drumming and vibration. Mounting feet shall be provided where necessary for bolting to a base or supports. Inlet and outlet shall terminate in flanges to facilitate removal. Stator vanes shall be of mild steel or aluminium alloy. The unit shall be designed to facilitate access to the impeller. Where motors are mounted external to the casing the requirements for drives and guards as mentioned elsewhere in this Specification shall be met. An access panel with purpose made air seal shall be provided in the fan casing, the access panel shall be sized to facilitate maintenance.
- (n) Mechanical Roof Extract Units
 - (i) The fans used in roof extract units shall meet the appropriate requirements of the preceding clauses relating to fans generally and to particular types of fans. The fan unit shall be self-draining. The motor and drive assemblies shall be installed with anti-vibration mountings. Cowls and bases shall be of materials, which are resistant to the weather, solar radiation and are appropriate to the location of the fan.
 - (ii) The impeller shall be manufactured of aluminium and is dynamically balanced, non-overloading and shall run in non-turbulent airflow.
 - (iii) Casings shall be formed to facilitate a weatherproof fitting to the building structure. Adequate access to electrical supply terminals and lubrication points shall be provided by means of hinged cowls or otherwise as appropriate. Back-draught dampers and/or fire release dampers shall be provided where indicated. Bird guards of not greater than 25 mm mesh shall be provided as an integral part of the unit and where no inlet ductwork is fitted, inlet protection guards shall be provided. An isolating switch shall be provided within the cowl for ease of maintenance.

- (o) Fans for Corrosive or Hazardous Applications
 - (i) Where fans are required to handle toxic, corrosive, flammable, explosive or high temperature gases the materials of construction shall be selected to be resistant to the gas being handled and all relevant safety regulations shall apply. Bearings and lubrication arrangements shall be suitable for the conditions. Fans shall meet the appropriate requirements of the preceding clauses relating to fans generally and to particular types of fans, the form of protection shall be as indicated. Where a protective coating is required to be used with corrosive gases the coating shall cover all parts of the complete fan, motor and casing assembly which will be in contact with the corrosive gases. No fan shall be installed if the protective coating has been damaged in any way. Impellers shall be of coated steel, stainless steel or aluminium or plastics as indicated. Where fans are installed in a potentially explosive atmosphere requirements are to be as indicated.

G7.2.5 Ductwork

- (a) General
 - (i) This Section contains requirements which, where relevant to this Contract, shall apply to the supply, delivery, handling, installation, jointing, testing and commissioning, cleaning of all ductwork, including ductwork insulation, where applicable, volume control dampers, non-return dampers, grilles, registers, diffusers etc.
 - (ii) Unless otherwise specified, the term "ductwork and fittings" when used in general context, shall mean all ductwork, fittings, dampers, supports, gauges, and all other components and accessories required for the complete installation of the system as shown on the Drawings.
 - (iii) The values of the air resistance to air flow within the ductwork and/or the total distribution system where indicated in the Specification or Drawing are only approximate. The Contractor shall verify these values based on the equipment offered and to provide a ductwork system that is capable of providing the specific air quantities to suit the space, temperature and noise level requirements.
 - (iv) On completion of the installation, all openings in the building structures etc. shall be made good and neatly finished by the Contractor.
 - (v) All structural and masonry dry-wall type air shafts shall be lined with the appropriate ductwork material to ensure no leakage of air occurs.
 - (vi) Ductwork shall be stiffened externally. Internal stiffeners shall not be acceptable. The Contractor shall ensure that no other services are suspended from the ductwork and fittings, unless otherwise approved by the S.O..

- (vii) The ductwork routing shall be designed by the Contractor, in combination with other equipment, piping etc. of the water treatment plant. The design air flow velocities shall not exceed the following:
 - Main air duct : 10 m/s
 - Branch Duct : 8 m/s
 - Duct outlets : 3 m/s
 - External louvres : 5 m/s
 - Grilles/registers/diffusers etc. : 2.5 m/s
- (b) Manufacture
 - (i) All ductwork and fittings delivered to site shall be new and shall be clearly and indelibly stamped to identify different materials, grades and manufacturers.
 - (ii) All ductwork shall be manufactured as far as possible to site dimensions as taken by the Contractor. Where site dimensions cannot be taken in advance, dimensions shall be taken from architectural/structural detailed dimensioned drawings. The Contractor shall make adequate provisions to accommodate any discrepancies that may occur between the Contract Drawings and site dimensions.
- (c) Ductwork and Fittings
 - (i) In general ductwork shall be manufactured from Galvanised mild steel sheet fabricated from 1.2 mm thick galvanised mild steel sheet unless otherwise specified. The material shall be an approved type for construction of ductwork. All ductwork and fittings shall be painted with primer and 2 coats of finishing paint after fabrication over the entire external surface area. Ductwork shall be true in section, without twists or distortion.
- (d) Dampers and Splitters
 - (i) General:
 - 1) The respective functions, types and general constructional requirements of dampers shall be in accordance with the latest ASHRAE guide unless otherwise indicated.
 - 2) Manually and automatically operated dampers shall include a means for indicating externally the position of the blades. Manual dampers shall include a device for positioning and locking the damper blades. The positions of all dampers as set after final regulation shall be indelibly marked at the adjusting device. Material used for the dampers shall be compatible with the ductwork material.
 - 3) Access openings with readily removable air sealed cover shall be provided adjacent to all dampers. Subject to limitations of duct size the dimensions of access openings shall not be less than 300 mm x 300 mm and they shall be located to afford access for inspection and maintenance.

- 4) All dampers shall be fitted with an adjustment mechanism of the approved type and shall be of rigid construction with no slack when opening or closing the dampers. The adjustment mechanism shall be concealed. The Contractor shall take special care to ensure that all the dampers are free from vibration and rattles under all operating conditions.
 - 5) Splitters shall be positioned as shown on the drawings or where necessary for the balancing the air distribution. Splitter damper shall be fitted with a hand level and quadrant in accessible positions and shall be capable of being locked in any position. The "OPEN" and "SHUT" positions shall be indelibly marked on the ductwork.
- (ii) Volume Control Dampers
- 1) Manually and automatically operated volume control dampers shall include a means for indicating externally the position of the blades. Manual volume control dampers shall include a device for positioning and locking the damper blades in position. The positions of all volume control dampers as set after final regulation shall be indelibly marked at the adjusting device for future references. Generally, fully open should be in line with the direction of flow.
 - 2) Access opening with readily removable air sealed covers shall be provided adjacent to all dampers. Subject to limitations of duct size the dimensions of access openings shall not be less than 300 mm x 300 mm and they shall be located to allow easy access for inspection and easy maintenance.
- (iii) Non-return dampers
- 1) Non-return dampers shall present a minimum resistance to airflow under running conditions and take up a stable position in operation. Maximum resistance shall be presented under reverse airflow conditions. Resilient strips or other purpose made devices shall be provided to prevent the damper rattling and to aid in air sealing under reverse flow conditions. They shall be designed for silent operation.
- (iv) Fire dampers
- 1) Fire dampers used singly or in combination shall have an overall fire rating not less than that indicated. In all cases evidence of fire rating in accordance with SS333 and shall be provided by an independent testing organisation in accordance to Fire Code 2018.
 - 2) Fire dampers shall be constructed of either a corrosion resistant material such as stainless steel or be galvanised or otherwise treated to minimise corrosion. The dampers shall be housed in a corrosion resistant casing constructed to avoid distortion due to stress in fire conditions. Provision shall be made to accommodate expansion of the damper blade(s) within the casing in fire conditions. A fire damper installation frame shall also incorporate provision for expansion within the surrounding structure together with lugs for building in to the structure.

- 3) Fire damper assemblies for installations in corrosive environments shall be fabricated from materials or be coated with a protective finish resistant to the corrosive substances and conditions indicated.
 - 4) Each fire damper casing shall be clearly marked with a permanent indication of the correct fixing altitude of the damper, the direction of air flow and the side at which the access/maintenance opening shall be located.
 - 5) The folded continuous interlocked blade type of damper may be used for vertical or horizontal duct applications. Multi-bladed dampers shall be provided with a means to ensure that all the blades close simultaneously.
 - 6) Gravity operated multi-bladed fire dampers shall not be used in vertical ducts.
 - 7) Gravity operated single blade dampers may be used for both vertical and horizontal ducts provided means are incorporated which ensure reliable and positive closure when operating in maximum air flow rate conditions.
 - 8) Where gravity acting off-centre pivoted dampers incorporate spindle bearings long term corrosion effects shall be minimised by choice of materials and bearings shall be sealed or capped to exclude dirt and dust. Damper blades shall close to comply with the stability and integrity requirements of BS 476 Part 8.
 - 9) For high velocity air systems fire dampers shall be of the type where damper blades, in the open position, give minimum interference with the airflow.
 - 10) Unless otherwise indicated, each fire damper shall be held in the open position by a corrosion resistant retaining device incorporating a fusible element.
 - 11) Fire dampers shall be located in such a position and be of a type which facilitate periodic manual release and re-setting for test purposes as per SS333 and Fire Code 2018.
- (v) Supports, Hangers and Brackets
- 1) All ductwork supports, hangers and brackets shall be designed to prevent strain in the ducts, transmission of noise, and allow easy removal of ductwork sections if required. Wherever possible all brackets shall be of the cradle or saddle type and shall be of aluminium alloy. For Galvanised steel ducts, galvanised steel supports are acceptable. Supports shall be provided at regular intervals. While hangers are acceptable to support the ductwork, additional supports shall be provided to prevent axial or lateral movement of the ductwork to the satisfaction of S.O..
 - 2) All horizontal ductwork shall be supported with aluminium alloy rods. Hanger rods brackets shall be of metal and shall have sufficient strength and durability and sufficient resistance to atmospheric corrosion. Hangers shall not be used in direct contact with dissimilar metals and shall be coated with aluminium or bitumastic paint.

- 3) Where angle stiffeners have been provided on the underside of the duct the hangers may pass through the ends of those angle stiffeners. Where this is not possible, angles for support for the underside of the duct shall be furnished and installed. The size of the angles shall be the same as that required for the stiffening angles for that particular duct section. All hanger rods shall be of minimum 6 mm diameter. Hanger rods shall be spaced as follows:

Cross-Sectional Perimeter of duct (m)	Maximum Spacing Between hangers (m)
Less than 1.5	2.4
1.5 to 2.5	1.8
Exceeding 2.5	1.2

- 4) Vertical ducts shall be securely supported by continuous lengths of structural angles of a size at least equivalent to that for stiffening. Angles fastened to opposite side of ducts shall extend across the openings and seat on the structure or slab on both sides of the openings.
- 5) Except as hereinafter provided ducts and all parts of the duct system shall be substantially supported and securely fastened to structural members of buildings with approved devices of non-combustible material, designed to carry the required loads. Use of expansion bolts in cinder concrete is prohibited. Connections shall not impair the effectiveness of the protection of structural members.
- (vi) Elbows, Bends and Branches:
- 1) Curved elbows shall have a centre line radius of not less than 1.5 times the radial width of the duct. Short radius vanned elbow shall be allowed only consequent upon structural or building requirements or limitations and only with the expressed approval of the S.O..
 - 2) Double thickness aerofoil turning vanes shall be provided for all right-angle elbows extending over at least 50 per cent of the whole curvature of the elbow. The turning vanes shall have a flange covering the whole base and shall be riveted to the duct at not more than 60 mm centres. Loosely or otherwise insecurely fitted vanes will be rejected.
 - 3) Where the radial width to depth ratio is less than unity a minimum of three (3) turning vanes shall be provided and in other cases, two vanes shall be provided. The double thickness aerofoil vanes shall be constructed of the same material of the duct enclosing them and shall be of two gauges heavier than the duct material.
 - 4) All branches shall be purpose made and designed to cause minimum loss and turbulence. In branches of greater than 300 mm duct width splitters shall be fitted. The splitters shall be hot-proof and fire-proof and shall be of resin bonded glass fibre materials or approved equivalent.

- 5) For FRP ducts, fittings shall be produced from the same resin system as the duct and conform to NBS PS 15-89. All fittings such as elbow, laterals, tees, and reducers shall be equal or superior in strength to the adjacent duct.
- (vii) Change in sections:
 - 1) Reductions in ductwork sections shall be made after any take off except that unless a reduction of 50 mm in the section can be made the original size shall be maintained. When a change section is made the slope of the reduction shall not exceed 1:4 for any side. Expansions in ductwork sections shall be made after any intake for the extracting air duct system. The slope of the expansion shall not exceed 1:7 on any side when a change in section is made.
- (viii) Joints
 - 1) General:
 - All joints shall be made substantially air-tight and no dust marks from air leaks shall be as shown at duct joints or connections to grilles, registers and louvres. The internal ends of all slip joints shall be made in the direction of the flow. All ducts shall be constructed with the fewest possible number of seams.
 - Anti-vibration joints shall be provided on either side of the fan duct sections and wherever necessary to minimise vibration, drumming or noise transmission through the ducts.
 - 2) Flexible joints:
 - Flexible joints shall be provided on fan inlet and outlet connections and elsewhere on ductwork where indicated or required. Joints shall be of the same cross-sectional area as the mating fan inlet, outlet or duct section. The centre lines of flexible joint connections shall be co-axial.
 - Flexible joints shall consist of, or be externally protected by, material having a fire penetration time of at least 15 minutes when tested in accordance with BS 476 Part 8 and shall comply with BS 476 Part 7, (Class 1: surface of very low flame spread).
 - 3) Joints for aluminium and galvanised mild steel ducts:
 - Longitudinal joints or seams shall be either continuous welded or lock formed, and sealer shall be used in the joints. For circular section ducts and smaller rectangular ducts in the width of the side a grooved seam shall be used. For rectangular ducts on the corner a Pittsburgh lock shall be used. All tails are to be formed to a tight lock.
 - On larger rectangular ducts and when extra rigidity is required a standing seam joint may be used and the joint shall be riveted at 150 mm centres. The joints shall be sealed.

- Transverse joints shall be either S, pocket, drive, reinforced bar slips or angle iron joints as shown on the following table.

Longer side mm	Type of Transverse Joint
Up to 300	S, drive or pocket slips at 2.2 m centres
310 to 450	S, drive 25 mm pocket 25 mm bar slips at 2.2 m centres
460 to 750 760 to 1050	38 x 38 x 4 mm angle iron joint or 38 mm bar slips with 25 x 4 mm bar reinforcing at 2.2 m centres
1060 to 1250 1260 to 1500	38 x 38 x 4 mm angle iron joint on 38 mm bar slips with 35 x 4 mm bar reinforcing at 2.2 m centres

4) Joints for FRP Ducts:

- Integral factory-made bell and spigot joints bolted or flanged joints shall be considered the standard means of joining duct sections and ducts to fittings. The minimum width of the overlay shall relate to wall thickness and shall be of the dimensions according to NBS PS 16-69. Duct and fittings shall be as crevice-free as is commercially practicable. All bolts and nuts used shall be stainless steel grade 316.
- Crevices between jointed pieces shall be filled with resin or thixotropic resin paste, leaving a smooth inner surface. All cut edges shall be coated with resin so that no glass fibres are exposed to the chemical environment.
- Exterior surface shall be smooth with no exposed fibres.

(ix) Access Doors

1) General:

- The Contractor shall provide access doors on all ductwork connections, either upstream or down stream of coils, upstream of fans, fire dampers, filters etc. to permit inspection, cleaning and maintenance purposes in accordance with SMACNA recommendations.
- Access doors shall be hung on approved heavy duty plated flat hinges and shall be secured in the closed positions by means of wedge type locks. All hinges shall be so positioned to enable the doors to remain open with additional supports. All access doors shall be hinged with two (2) hinges and two (2) wedge type locks for doors of dimensions above 350 mm x 350 mm. Hinges shall be strong and robust and capable of withstanding any abnormal loading which may be imposed during the opening or closing of the access doors without visible strain or deformation.

- Where it is impractical to use hinged doors, the access doors shall be fixed in position with wedge type locks on two opposite sides with a minimum of four (4) numbers of locks per door.
- Access doors for aluminium ductwork shall be of double panel welded construction. The door panel shall not be less than 1.6 mm aluminium sheet and shall be provided with above 12 mm wide neoprene rubber gasket around the entire perimeter. The access doors shall form an internal part of the ductwork and be of the same fire rating as the ductwork system.
- Access doors for galvanised steel ductwork shall be of double skin construction, with insulation where applicable, and shall be of the same general character as the adjacent sections of ductwork. The door panel shall be constructed from sheet metal with thickness of not less than 1.0 mm and shall be provided with neoprene rubber gasket of above 12.0 mm thickness around their entire perimeter. The access door shall form an internal part of the ductwork and be of the same fire rating as the ductwork system.

(x) Flexible Ductwork

- 1) Where flexible ductwork is indicated, or required between rigid ductwork and items of equipment, the internal diameter of the flexible duct shall be equal to the external diameter of the rigid duct and equipment spigot. Flexible ductwork shall be kept to a minimum and shall not be used between rigid sections of ductwork to change direction unless indicated or approved by the S.O.. The maximum length of any individual flexible duct shall not exceed 1.0 m unless otherwise indicated or approved by the S.O..
- 2) The flexible duct shall have a liner and a cover of tough tear-resistant fabric equal in durability and flexibility to glass fibre fabric. The fabric shall be impregnated and coated with plastics. It shall be reinforced with a bonded galvanised spring steel, stainless steel or other approved wire helix between the liner and the cover. An outer helix of glass fibre cord or equal shall be bonded to the cover to ensure regular convolutions. Flexible ductwork without a liner may be used subject to compliance with all other appropriate requirements of this Section.
- 3) Alternatively, flexible ductwork shall consist of flexible corrugated metal tubing of stainless steel or aluminium. The metal surface(s) may be coated with a plastic material.
- 4) Joints at flexible duct connections shall be made with a sealant which permanently retains adhesion and elasticity through the design working temperature range. The sealant and method of application shall be in accordance with the ductwork manufacturer's recommendations. Flexible ductwork up to and including 140 mm diameter shall be secured with a worm drive type hose clip complying with BS 5315. Ductwork over 140 mm diameter shall be secured with a band clip.

- 5) The frictional resistance to air flow per unit length of flexible duct shall not exceed 150% of the frictional resistance per unit length of galvanised steel duct of similar diameter. The radius and D is the diameter of the flexible duct.
 - 6) Flexible ductwork shall be suitable for an operating temperature range of -50°C to +900°C and shall comply with BS 476 Part 5, Rating Class P; Part 6 having an index of performance (I) not exceeding 12 of which not more than 6 should derive from the initial period of test; Part 7 Class 1 (surface of very low flame spread) unless otherwise indicated.
- (xi) Fire Rated Ductwork and Fittings
- 1) Ductwork in areas which are designated as fire compartments or which form part of an engineered smoke control system shall be fire rated to give a minimum of one (1) hour fire rating inside and one (1) hour fire rating outside the ductwork.
 - 2) Fire rated ductwork shall be formed from minimum 1.2 mm thick galvanised sheet metal, with all seams welded and all connection fully completed with gasket. All seams shall be cold galvanised after welding to make good the surface protection. The duct shall be surrounded by a fire rated material which shall be approved for use by the relevant Authority. The fire rated ductwork construction shall also be subjected to the approval of the relevant Authorities.
 - 3) All necessary cover strips, aligning strips and other accessories required for the complete installation of the fire rated ductwork shall be supplied by the same manufacturer as the sheet metal.
- (xii) Air Terminal Devices:
- 1) General:
 - The testing and rating of air terminal devices and the definition of terms used concerning such devices shall be in accordance with BS EN ISO 5135.
 - The sizes of all air terminal devices shall be based on the dimensions indicated and shall be suitable for the air volume flow rate, air diffusion and noise rating indicated.
 - The materials of construction shall be galvanised steel, aluminium or plastics as indicated. All items shall be protected against corrosion and be provided in fully finished condition.
 - 2) Grilles and registers:
 - All grilles, louvers and registers shall be selected for quiet operation when handling the specified volume of air and shall be arranged to provide uniform velocity across the face of the grilles, louvers and registers.
 - The air quantities as shown on the Drawings are only notional. The Contractor shall ensure that the air quantities provided through the grilles and registers are adequate to suit the site space, temperature and noise level requirements.

- Each supply air grille (SAG) and register (SAR) shall have two sets of separately adjustable blades, one set horizontal and one set vertical. Each supply air register shall, in addition, be provided with an airflow rate controller, which shall be an opposed blade multi-leaf type unless otherwise indicated. Airflow rate controllers shall be adjustable from the front of the grille or register. For up to ten grilles or registers one set of tools required for adjustment shall be provided; from eleven to twenty-four grilles or registers, two sets; twenty-five and above numbers of grilles or registers, three sets.
- Each exhaust air grille (EAG) may have either a single set of blades or bars, either vertical or horizontal, or a lattice or egg crate front, or shall be as indicated. Each exhaust air register (EAR) shall, in addition, be provided with an airflow rate controller which shall be an opposed blade multi-leaf damper operable from the front of the register.
- All grilles and registers shall be from the product range of reputable manufacturers and their performance shall be tested and certified. Full performance data and samples for each different type of grille or register shall be submitted for the approval of the S.O..

3) Diffusers:

- Diffusers shall be of stainless steel, aluminium or plastics as indicated. Diffusers mounted on ceilings shall have anti-smudge surrounds. Pan type diffusers may be provided except where cone type diffusers are indicated.
- Each diffuser shall be provided with an air flow rate controller. Where a diffuser is directly connected to a stub duct which has a straight length of less than two diameters or equivalent then an equalising deflector shall be used.

4) External Louvres and Insect Screen:

- All external louvres for air supply intake and air exhaust outlet shall be, unless otherwise specified, provided by the Contractor to the approval of the S.O.. The Contractor shall also provide insect screens behind all external louvres.
- Louvre blades shall be horizontal. Louvres (blades and frame) shall be parallel constructed of anodised aluminium of the same colour as the windows provided for the buildings. Colour shall be by pigmentation introduced during the anodising process. Blades shall be set approximately 50 mm apart and of an angle of 300 to the horizontal. The louvres shall be of the weatherproof type.
- All supply air louvres (SAL) and exhaust air louvres (EAL) and openings shall be fitted with 5 mm mesh matt back painted insect screen in between the external louvres and the ductwork. The installation shall be such that the insect screen can be easily removed for periodic cleaning.
- All external louvres shall be designed with more than 50% effective free area at velocity limit specified based on the free air space.

(xiii) Vacuum Relief Valves

- 1) The air handling ductwork and cover systems shall be provided with adequate vacuum relief valves or other equivalent devices to limit the maximum negative pressure within the ductwork system and beneath the covers or aluminium domes to less than 1.0 kN/m² under the most onerous conditions of ventilation fans operating at full capacity.

(xiv) Air Filter

- 1) Each supply air fan (SAF) shall be arranged with an approved free flow air filter adjoining the duct inlet. The filters shall be easily accessible and arranged for easy cleaning or changing of the filter elements. All filters must have Minimum Efficiency Reporting Value (MERV) >6.
- 2) The filter medium shall synthetic fibre material. The filter shall have minimum average arrestance of not less than 85% by ASHRAE 52-68 Gravimetric Test. The initial and final resistance of the media shall not exceed 40 Pa and 180 Pa respectively both at 100 m/min air velocity.

(xv) Test Holes

- 1) The Contractor shall provide all test holes in ducts for testing by means of volumeter anemostat or Pitot tube at all branches and other positions called for by the S.O. to enable a comprehensive selection of velocity readings to be taken. All proposed positions of test holes shall be indicated on the detail drawings.
- 2) Test holes shall be factory fabricated, airtight, non-corrosive and complete with screw cap and gasket.
- 3) The test holes shall be at 150 mm or less centres across the longer side of the duct and shall consist of a 25 mm tap screw brazed into a 40 mm x 5 mm bar riveted to the duct. Each tap screw shall be fitted with a plug.
- 4) Where test holes are fitted to the internally insulated ductwork, the ductwork insulation shall be protected with perforated metal internally for a distance of 600 mm upstream and for 150 mm downstream of the test hole position.

(xvi) Noise and Vibration Control

- 1) Wherever possible noise shall be contained at its source. Approved packaged sound absorbers such as attenuators, silencers, mufflers, etc. shall be fitted immediately adjacent to the fan so that the sound power is reduced at source before reaching the distribution system. Additional attenuation may be provided in the distribution ductwork by lining some lengths of ducts, or preferably bends with acoustic absorbent material.
- 2) The fan sound pressure level, measured under installed conditions at a distance of three fan diameters from the fan centre, shall not exceed 85 dB(A). If necessary, a silencer may be provided and the cost shall be deemed to be included in the tender sum.

(xvii) Finishes

- 1) All supports, hangers, brackets, stiffeners etc. shall be painted with one coat of primer before erection and shall be further painted after installation with 2 coats of finishing paint, colour to S.O.'s direction.
- 2) All dampers and other equipment forming part of the ductwork system or connected to the ductwork shall be painted with primer and 2 coats of finishing paint, colour to match duct.
- 3) Any damage to factory applied finishes shall be made good on site in the manner recommended by the manufacturer and to the satisfaction of the S.O..

(xviii) Cleaning of Ductwork

- 1) During installation, the Contractor shall provide properly fitted covers shall be provided at all times to prevent ingress of rubbish. Rags, newspaper and like materials will not be accepted as suitable closures.
- 2) Should the existence of rubbish be found in air ducts, the ductwork shall be dismantled as necessary to ensure complete removal of rubbish. All ducts shall be cleaned of dust before starting of fans.

(xix) Testing and Commissioning

- 1) Contractor is to submit T & C procedure and schedule for S.O.'s approval.
- 2) On completion of the work, the Contractor shall supply all necessary calibrated instruments, facilities, labour, equipment, materials etc for the measurements of air velocity and pressures within the ducts and through the louvres.
- 3) All ductworks are to be tested in sections on completion of each section.
- 4) The Contractor shall test and demonstrate in the presence of the S.O. that the specified air capacity of the fans (or the number of air changes per hour) is achieved and that the air distribution between louvred inlets or extracts is evenly and properly distributed in accordance with the approved flow distribution calculations.
- 5) To determine the amount of airflow through ducts or louvres, an average velocity of airflow shall be used. This average velocity shall be calculated as the arithmetic mean of various point velocities in the duct.
- 6) All the equipment shall be run and shown to perform satisfactorily in all respects under a full load working test of not less than 24 hours duration.
- 7) All Inspection and T & C results are to be submitted to S.O. after the completion of the inspection and T & C. once verified, these are to be incorporated in the OMM.
- 8) Defects in materials, workmanship or performance of the ventilation systems shall be rectified by the Contractor.
- 9) After rectification, T & C is to be carried out again.

END OF SECTION